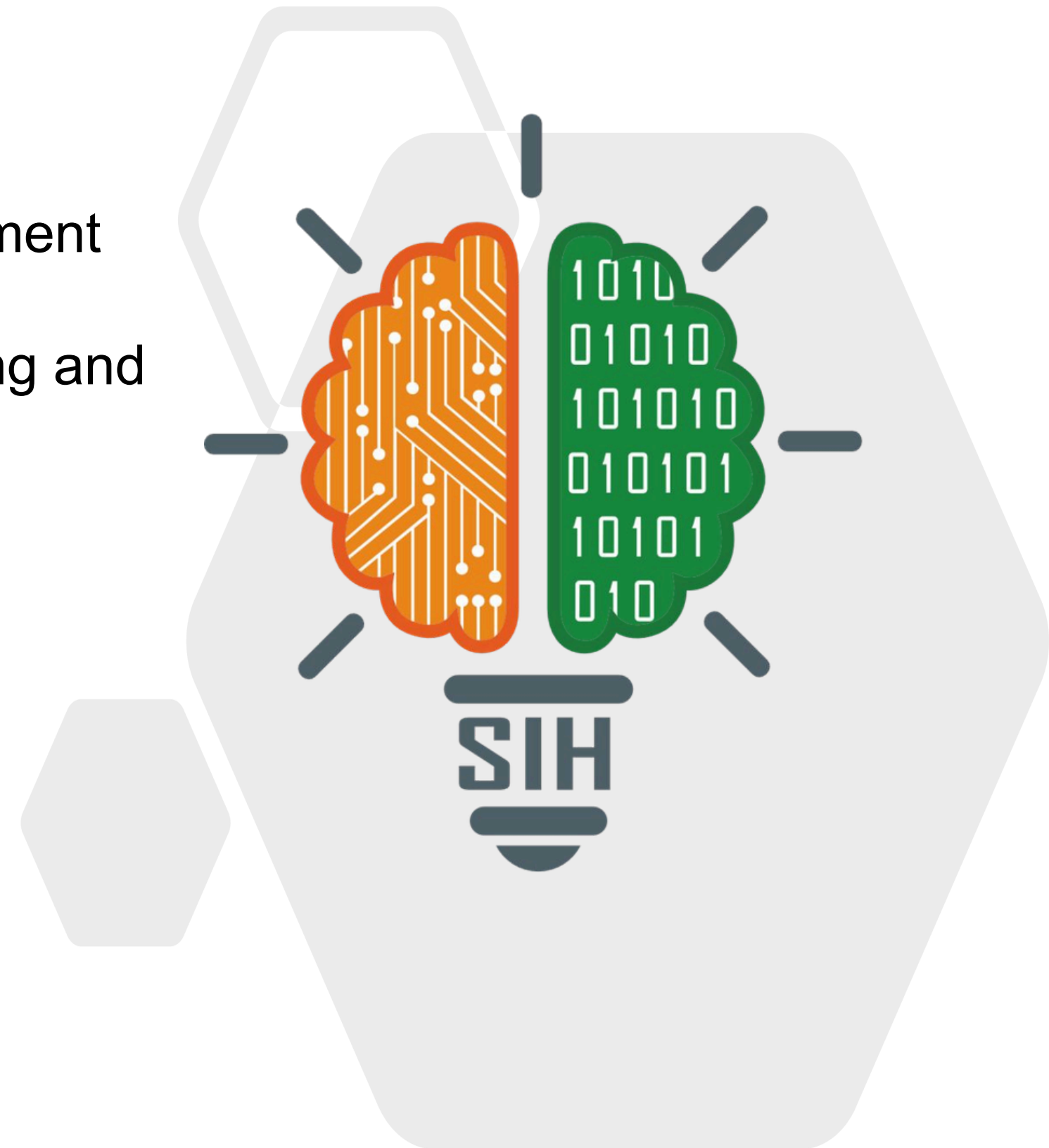


SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

- **Problem Statement ID** - 26206
- **Problem Statement Title** - Disaster management includes ideas related to risk mitigation, Planning and management before, after or during a disaster
- **Theme**- Disaster Management
- **PS Category**- Software
- **Team ID**-
- **Team Name** - Team Sinchan



AEGIS, an AI decision-support layer that sees, prioritizes, and routes disaster response

THE PROBLEM

FLOODS ARE DISRUPTING LIVES ACROSS INDIA

In 2026, extreme rainfall and flooding have affected multiple states, displacing thousands of families, damaging infrastructure and overwhelming response systems.

- **Large and rapidly changing impact**
Extent of damage is often unclear and varies across areas.
- **Difficult prioritization**
Critical areas are hard to identify with limited real-time information.
- **Inaccessible infrastructure**
Floods and debris make many roads and bridges unusable.
- **Communication gaps**
Field teams face challenges in sharing timely updates.
- **Time critical decisions**
Decisions must be made quickly, with limited resources.



OUR SOLUTION

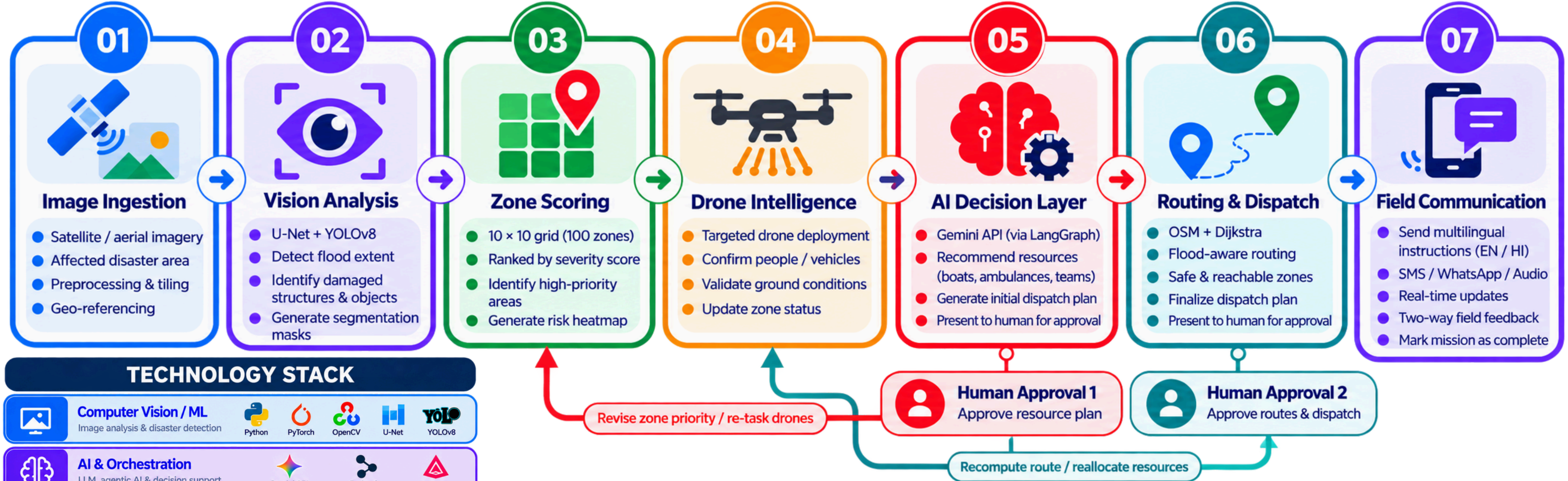
A UNIFIED, AI-POWERED RESPONSE SYSTEM

AEGIS brings together satellite data, AI analysis and real-time information to see the disaster, prioritize critical zones, plan resources and route rescue teams while keeping humans in the loop.

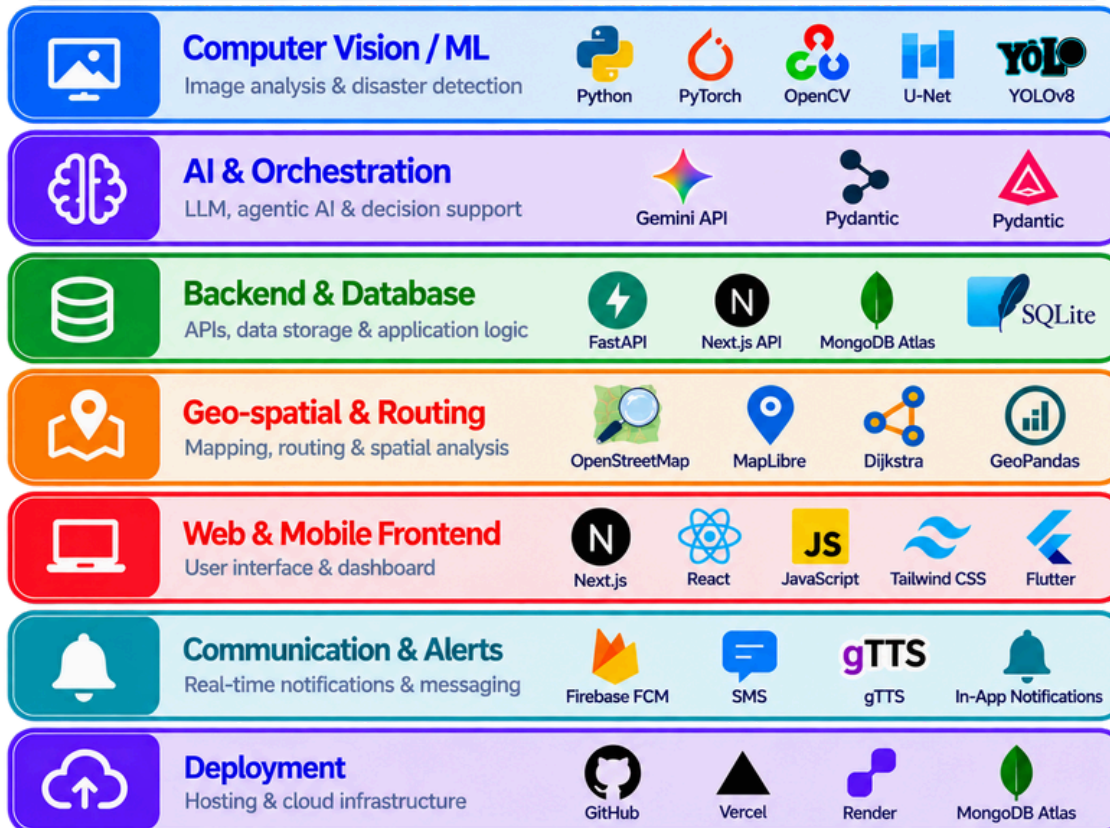
- **01 Vision Analysis**
Understand what is happening
Satellite and aerial images
U-Net for flood segmentation
YOLOv8 for affected areas
- **02 Severity Prioritization**
Find what needs help first
Flood extent, damage score, population density
100-zone grid
Priority zone selection
- **03 Drone Intelligence**
Get real time ground insights
Targeted drone deployment
Verify conditions and update zone status
- **04 Decision Layer**
Recommend the best actions
Gemini AI for resource allocation
Route planning and risk assessment
Human in the-loop approval
- **05 Dispatch**
Turn plans into action
Optimized flood-aware routing
OSM and Dijkstra
SMS and WhatsApp notifications

TECHNICAL APPROACH

End-to-End AI Pipeline, From Disaster Image to Emergency Dispatch



TECHNOLOGY STACK



100 Zones made by AEGIS to find critical areas (approximation)



Water Coverage by using U-NET



A practical, scalable approach to turn disaster information into faster, safer, coordinated rescue action.



FEASIBILITY

- Prototype validated end-to-end on real satellite imagery and open datasets (e.g., FloodNet, RescueNet).
- Built using widely available tools and open-source technologies.
- Can operate with minimum hardware and internet requirements, making it deployable in resource-constrained regions.
- Designed with human-in-the-loop control for practical, on-ground decision-making.



VIABILITY

- Reduces the time and effort required to identify priorities and coordinate resources.
- Can be integrated with existing disaster management systems and government workflows.
- Modular architecture allows gradual scaling from pilot deployment to state/national level.
- Revenue and sustainability through government adoption, partnerships, and open-source community support.



SUSTAINABILITY & SCALE

- Starts with a focused use case (floods) and can extend to other disasters (earthquakes, landslides, cyclones).
- Leverages cloud infrastructure (Vercel, Render, MongoDB Atlas) for scalable and reliable deployment.
- Works alongside existing rescue teams, authorities and local systems.
- Designed for long-term use with continuous updates, community feedback and multi-region support.



Multi-Agents

Faster Coordination

Reduction in response time compared to traditional methods



100

Zones Scored

10 × 10 grid, ranked by severity and resource need



2

Human Approval Gates

Checkpoints before any dispatch action

WHY THIS SOLUTION IS PRACTICAL

1

Detect

Identify affected areas early



2

Prioritize

Focus on the most critical zones



3

Coordinate

Match resources to actual need



4

Respond

Guide teams with safer, optimized routes

IMPACT AND BENEFITS

IMPACT: FASTER DECISIONS • BETTER RESOURCE USE • SAFER RESCUE • MORE TARGETED RELIEF

HOW AEGIS CREATES IMPACT

01

Early identification

Large, affected areas received quick detection through satellite imagery and AI analysis.

02

Priority-based action

Response starts with the zones showing the greatest need, not equal treatment.

03

Need-based deployment

Available resources are matched to the actual situation in each zone.

04

Safer movement

Planned routes account for flooded or hazardous areas to avoid delays.

05

Clear communication

Field teams receive simple, multilingual instructions and real-time updates.

KEY BENEFITS



Faster emergency response

From data to action, in minutes.

- Reduces the time needed to identify the most affected zones.
- Faster decision-making for routing, deployment and alerts.
- Improves the first response from observation to action in minutes.



Better use of limited resources

Right help, right place, right time.

- Prioritizes boats, ambulances, and teams where they are actually needed.
- Avoids sending the same type or amount of help everywhere.
- Supports planning when resources are limited or scarce.



Safer rescue operations

Smarter routes, safer teams.

- Creates flood-aware and safe routes for teams on the ground.
- Helps teams avoid high-risk or inaccessible areas.
- Reduces exposure to hazards and improves on-ground safety.



More targeted relief & coordination

Unified view, greater impact.

- Combines data from multiple sources for a complete view.
- Helps authorities and field teams work with the same real-time information.
- Ensures aid reaches the right people in the right areas.

WHO BENEFITS?



Authorities

Better situational awareness, clearer priorities, faster coordination.



Rescue Teams

Safe routes, real-time updates, more efficient operations.



Affected People

Faster assistance, targeted relief, safer communities.

Flood & Disaster-Damage Datasets

- [xBD — building-damage pairs, 19 events](#)
- [FloodNet — Harvey UAV flood masks](#)
- [RescueNet — UAV post-disaster segmentation](#)
- [Sen1Floods11 — Sentinel flood-extent labels](#)

Core Computer-Vision Models

- [U-Net — flood-extent segmentation](#)
- [YOLOv8 — damage & object detection](#)
- OpenCV, PyTorch, NumPy — pipeline tooling

AI & Orchestration

- [Google Gemini API — resource-decision reasoning](#)
- [LangGraph — agent orchestration](#)

Satellite & Geospatial Sources (India)

- [Bhuvan / NRSC — ISRO geo-portal + flood warning](#)
- [Bhoonidhi — ISRO EO data archive](#)
- [MOSDAC — ISRO met/ocean satellite data](#)
- [VEDAS — ISRO flood-mapping platform](#)

Global / Open EO Sources

- [Copernicus Data Space Sentinel-1/2 imagery](#)
- [USGS EarthExplorer, Landsat archive](#)
- [NASA FIRMS, hotspot layers, multi-hazard](#)

Geo-spatial & Routing

- [OpenStreetMap, base road network](#)
- [NetworkX, Dijkstra safe-path routing](#)

Existing Govt. Platform, we can integrate

- [Sachet, NDMA early-warning app](#)
- IDRN, India Disaster Resource Network
- NDEM, Nat'l Database for Emergency Mgmt
- C-FLOOD, CWC unified flood forecasting (2026)
- Nat'l Flood Vulnerability Assessment System, Bhuvan/NRSC
- Aapda Mitra, NDMA volunteer scheme

Standards & Institutional Frameworks

- Disaster Management Act, 2005, NDMA/SDMA/DDMA
- [UNDRR, Sendai Framework 2015-2030](#)
- [Copernicus EMS Rapid Mapping](#)

Data & Storage

- SQLite, JSON, Pandas, GeoPandas, persistence + audit trail

Communication & Field Delivery

- [Twilio API, SMS/WhatsApp dispatch](#)
- [Google Text-to-Speech, audio instructions](#)
- Google Translate API, multilingual delivery

Mobile & Offline Resilience

- [Flutter, field-responder app](#)
- Firebase Cloud Messaging, push alerts
- Hive, offline route caching

Command-Centre Frontend

- [Next.js / React, dashboard framework](#)
- [Leaflet, zone-grid map & heatmaps](#)